BI-DIRECTIONAL BEVERAGE ANTENNA 1 COAX OUTPUT MANUAL

Bi-Directional TWO-WIRE BEVERAGE TWO WIRES
SWITCHABLE DIRECTION 50 Ω COAX
HIGH 1.5 - 2.5 m
SPACING 10 - 20 cm
DIAMETER 0.3 - 1 mm
COAX
TERMINATOR
manual
RX ANT

With careful attention to design and construction, a single beverage antenna can be made to cover two directions with good performance. The transformer at the end of the beverage is designed to match impedance of the antenna. Compare to single wire beverage antenna, where there is only load resistor, in this system all transformers must be calculated exactly for the antenna design (height, wires space and diameter etc).

Jump to

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- **Terminator Box**
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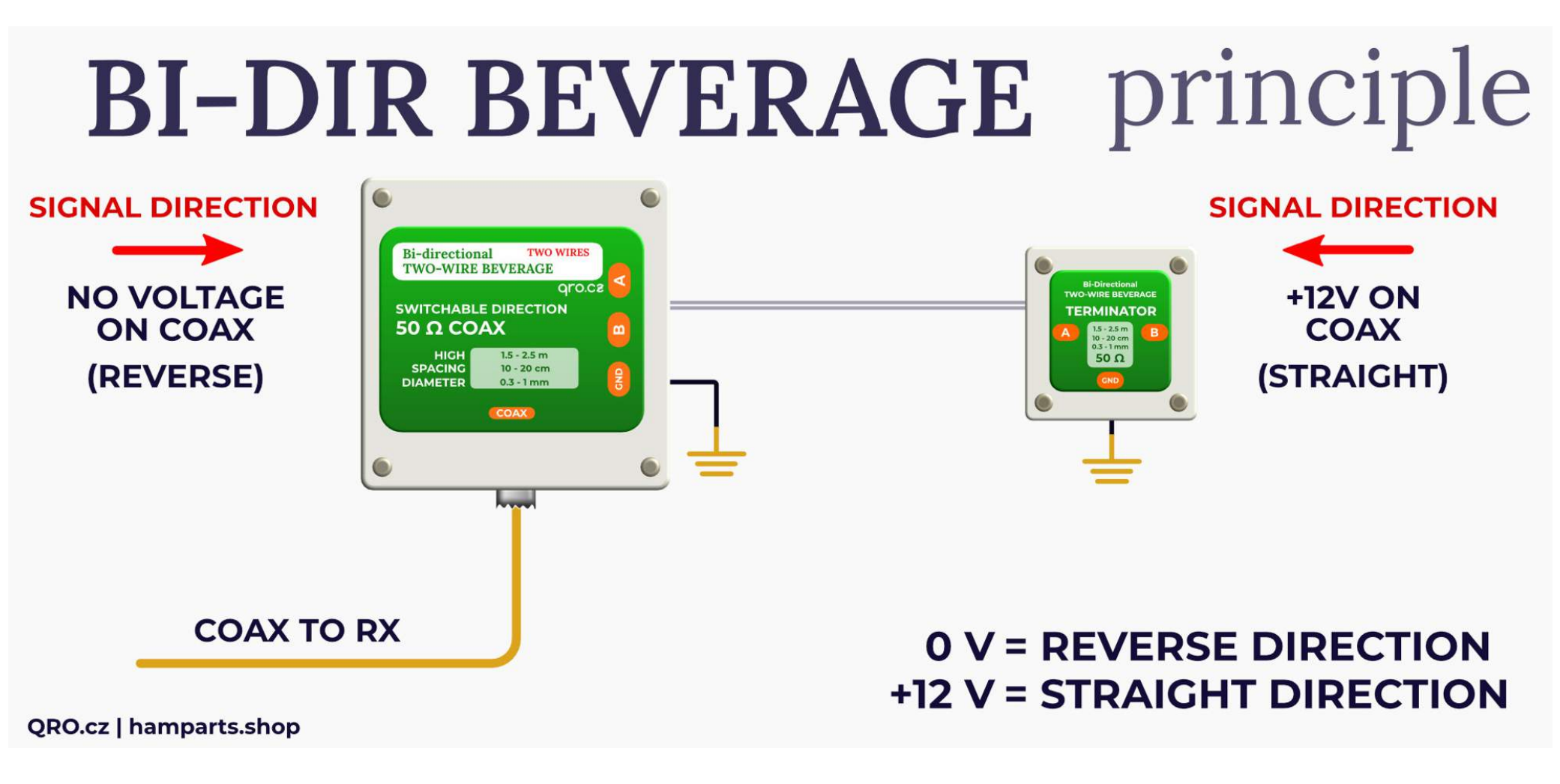
Principle

This one coax output version has got internal relay for direction switching. Control of the relay is over the coax. Without voltage the antenna has REVERSE direction (compare to single wire construction). With voltage on coax you switch to STRAIGHT direction (the same as on single wire beverage).


BI-DIR BEVERAGE principle
SIGNAL DIRECTION
NO VOLTAGE ON COAX (REVERSE)
SIGNAL DIRECTION
+12V ON COAX (STRAIGHT)
COAX TO RX
0 V = REVERSE DIRECTION
+12 V = STRAIGHT DIRECTION
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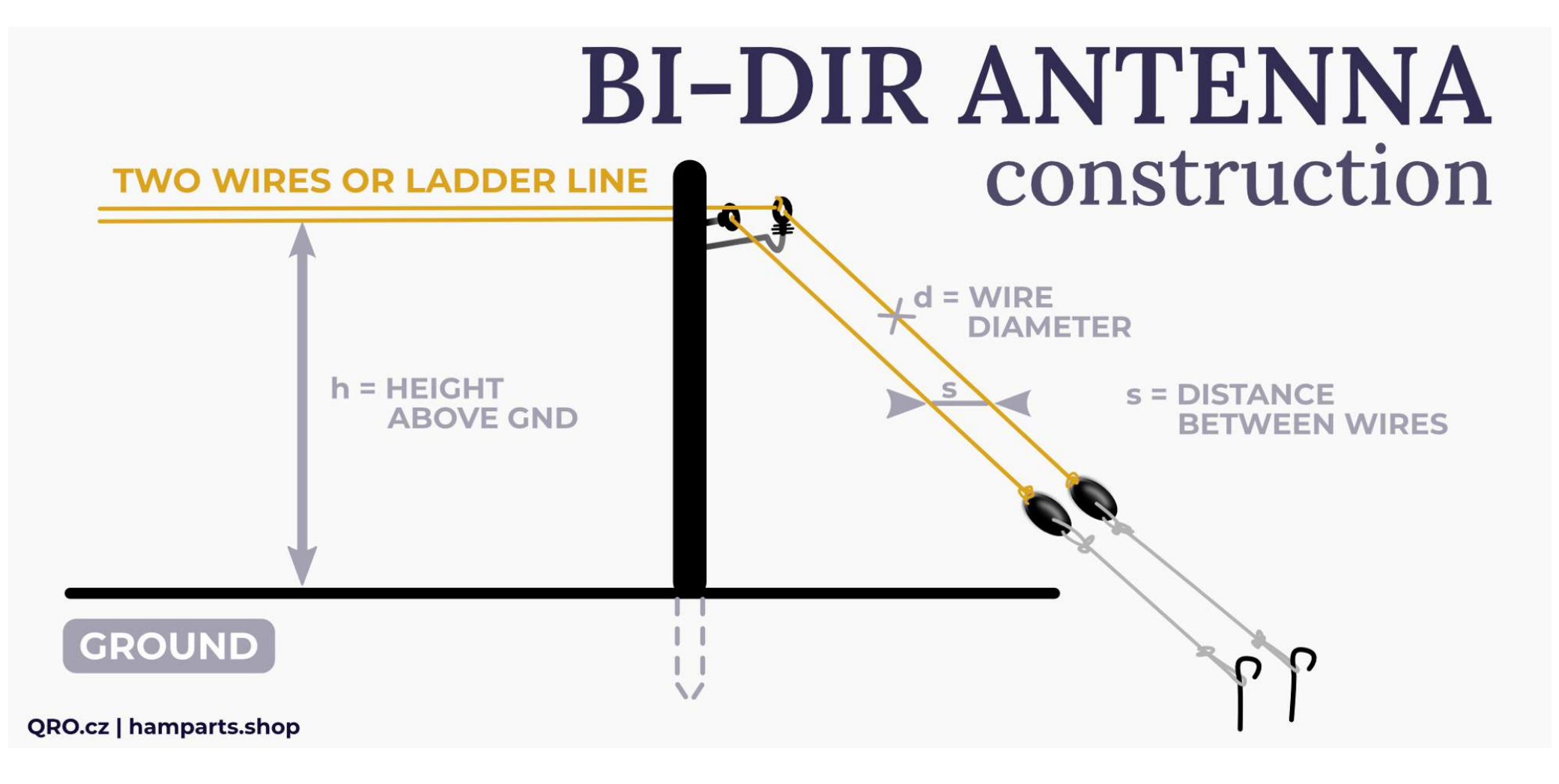
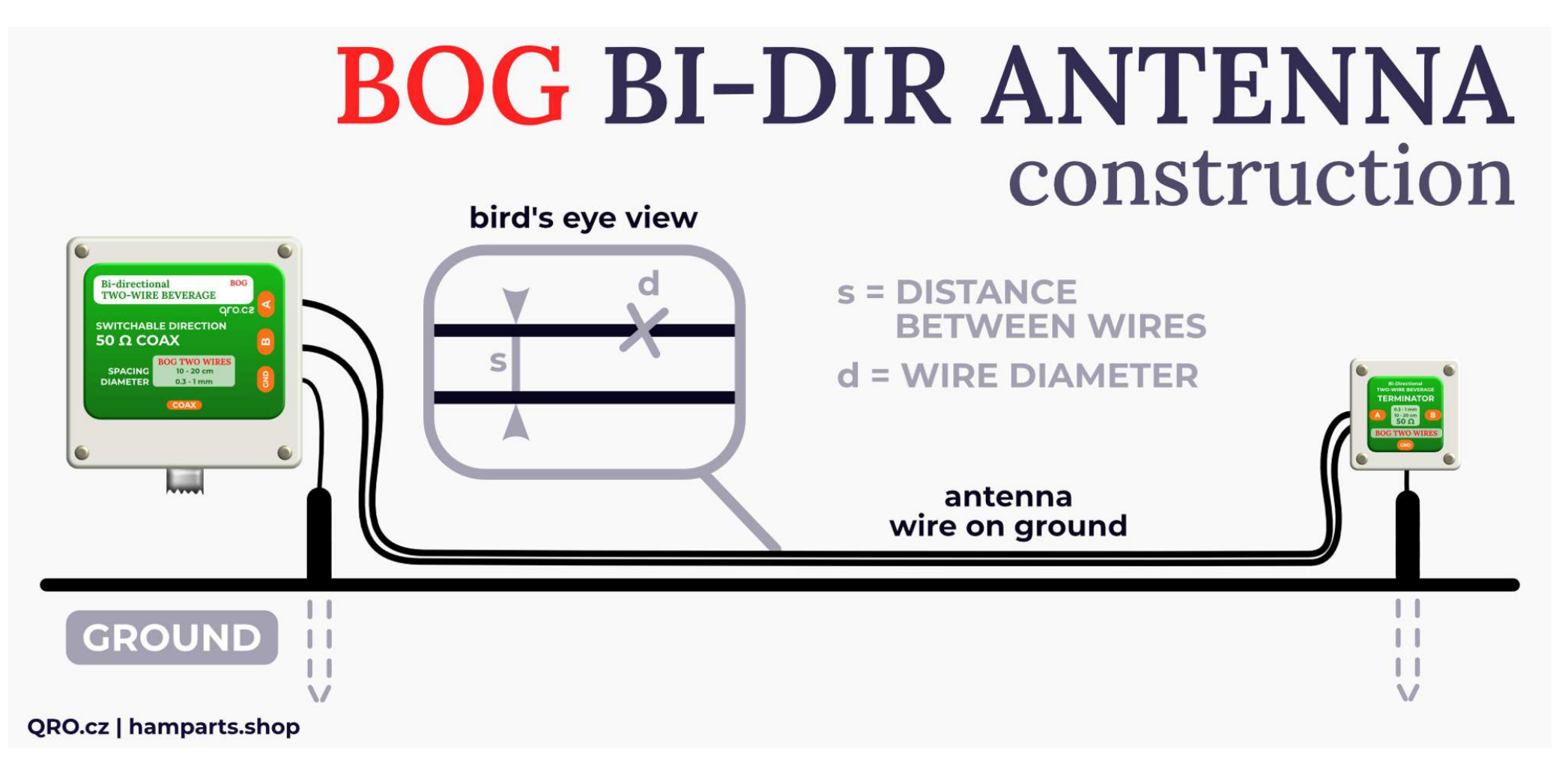
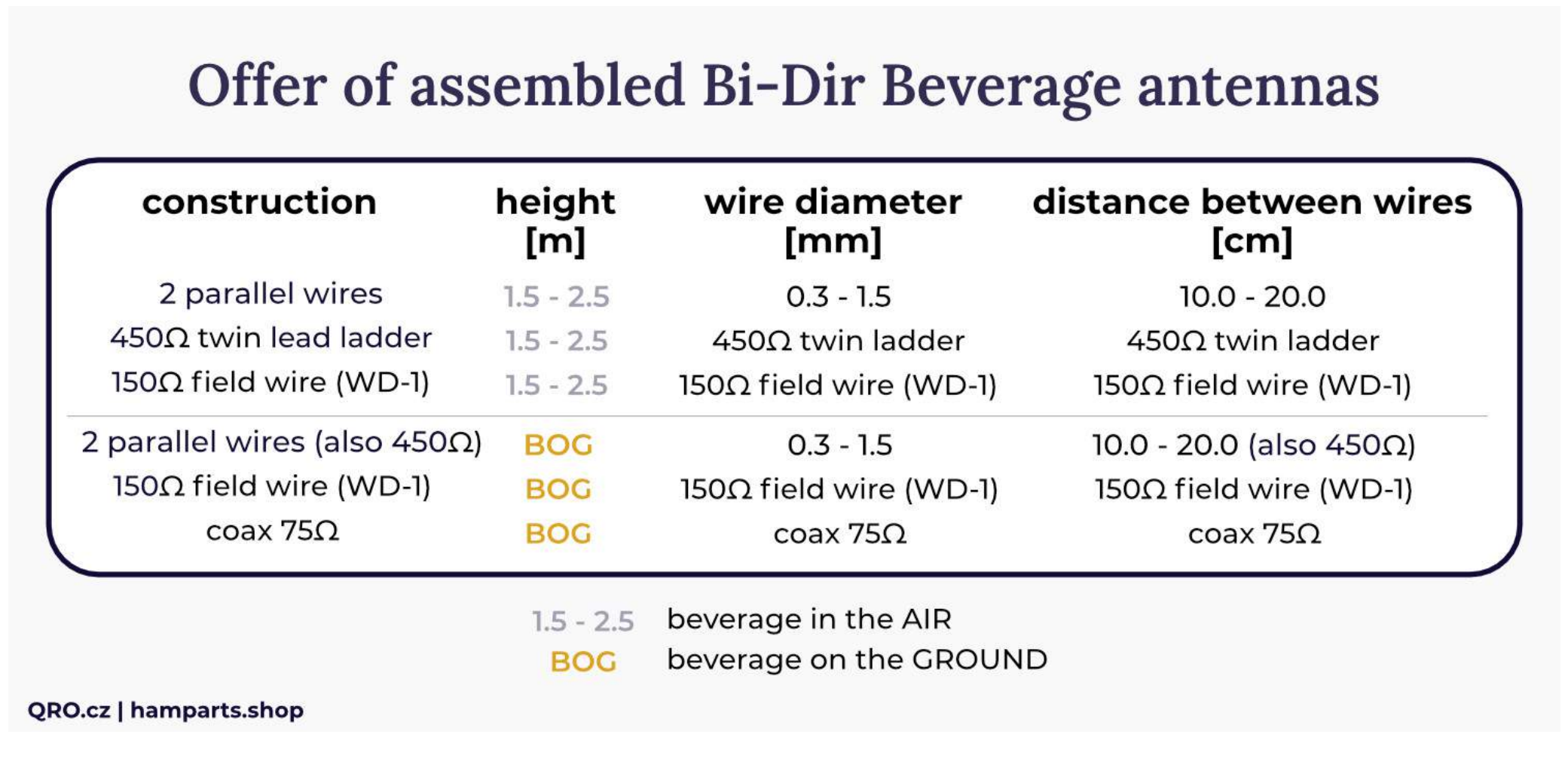
Construction

Beverage antennas should be installed more than a wavelength from other large metallic structures, such as vertical antennas, towers, power lines, electric fence wires etc. Antenna units can be mounted on trees, wooden pole or fiberglass rods - **DO NOT use metal pipes and never mount this unit with the connectors facing up**.

The parallel wire line used to construct the antenna must be perfect balanced transmission line conditions. The wires must be parallel to the ground. A line should not be constructed in a vertical fashion, using single support poles with one wire under the other. The closer the wire spacing of the line, the lower the impedance of the line, and the less critical installation-related unbalances become. When the antenna is ladder-line (300 or 450 Ohm) - twist the ladder line multiple times along its length.


BI-DIR ANTENNA construction
TWO WIRES OR LADDER LINE
h = HEIGHT ABOVE GND
d = WIRE DIAMETER
s
s = DISTANCE BETWEEN WIRES
GROUND
QRO.cz | hamparts.shop


BOG BI-DIR ANTENNA construction
bird's eye view
s
d
s = DISTANCE BETWEEN WIRES
d = WIRE DIAMETER
antenna wire on ground
GROUND
QRO.cz | hamparts.shop



Offer of assembled Bi-Dir Beverage antennas

construction	height [m]	wire diameter [mm]	distance between wires [cm]
2 parallel wires	1.5 - 2.5	0.3 - 1.5	10.0 - 20.0
450Ω twin lead ladder	1.5 - 2.5	450Ω twin ladder	450Ω twin ladder
150Ω field wire (WD-1)	1.5 - 2.5	150Ω field wire (WD-1)	150Ω field wire (WD-1)
2 parallel wires (also 450Ω)	BOG	0.3 - 1.5	10.0 - 20.0 (also 450Ω)
150Ω field wire (WD-1)	BOG	150Ω field wire (WD-1)	150Ω field wire (WD-1)
coax 75Ω	BOG	coax 75Ω	coax 75Ω

1.5 - 2.5 beverage in the AIR
BOG beverage on the GROUND

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Feeder Box

This box contains transformers, direction relay, common-mode filter and protections.

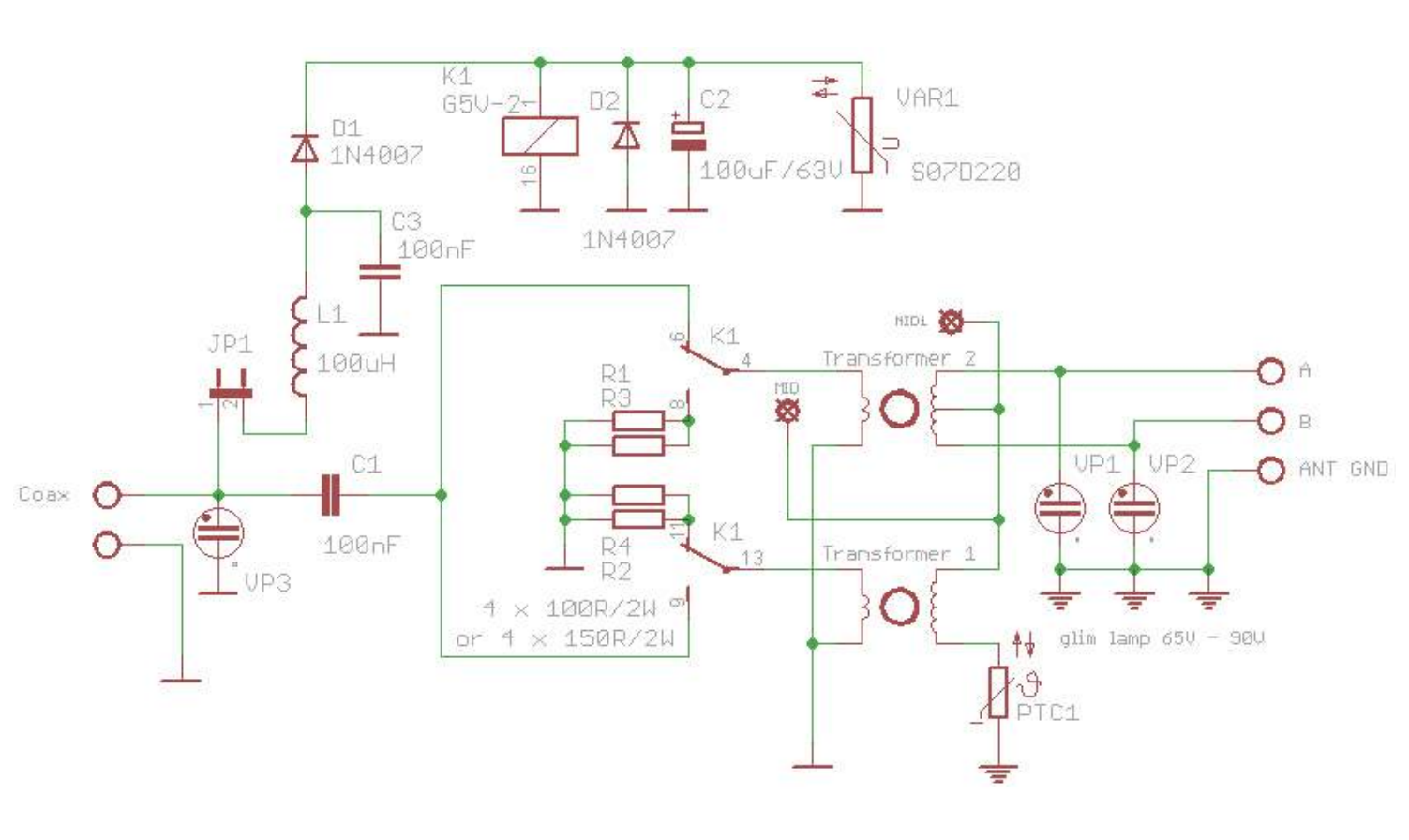
1 coax version with PL SO-239 connector

Bi-Directional TWO-WIRE BEVERAGE TWO WIRES
SWITCHABLE DIRECTION 50 Ω COAX
HIGH 1.5 - 2.5 m
SPACING 10 - 20 cm
DIAMETER 0.3 - 1 mm
COAX

1 coax version with F connector

Bi-Directional TWO-WIRE BEVERAGE
SWITCHABLE DIRECTION 75 Ω COAX
450 Ω LINE
HIGH 1.5 - 2.5 m
COAX

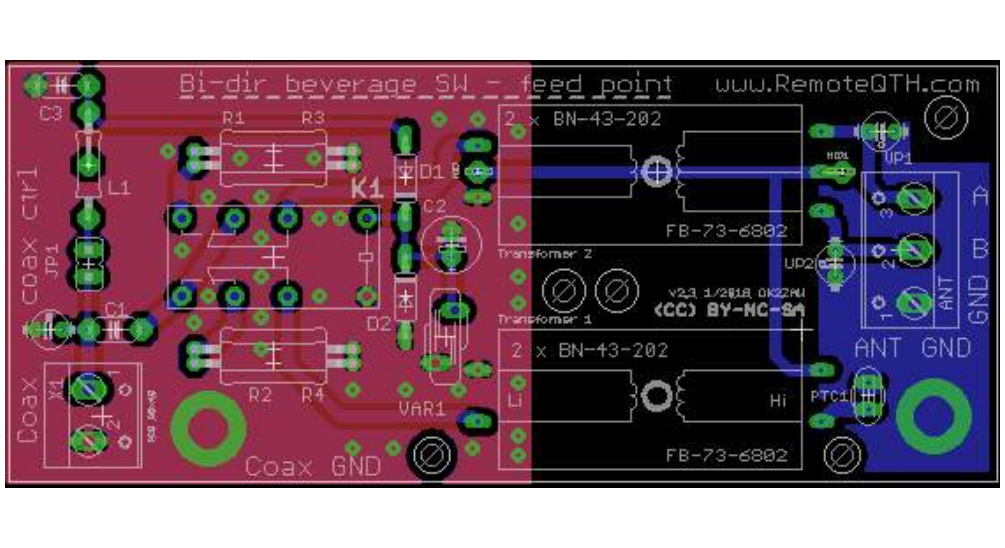
Schematic


K1 G5V-2
D1 1N4007
D2
C2 100uF/63V
VAR1 S07D220
1N4007
C3 100nF
L1 100uH
JP1
Coax
C1 100nF
VP3
R1 R3
R4 R2
4 x 100R/2W or 4 x 150R/2W
K1
Transformer 2
Transformer 1
VP1 VP2
A
B
ANT GND
glim lamp 65V - 90V
PTC1

Part list

NAME	PART	VALUE
VP 1-3	neon lamp	lamp 65V
C1, C3	capacitor	100nF 100V
C2	capacitor	100uF 50V (63V)
D1, D2	diode	1N4007
R1 - R4	resistor	100Ω/2W (50Ω) 150Ω/2W (75Ω)
PTC1	PTC	PTC 100mA
VAR1	varistor	14V DC
K1	relay	12V
Tr1, Tr2	transformer	BN-6802-73

PCB


Bi-dir beverage SW - feed point
www.RemoteQTH.com
2 x BN-43-202
FB-73-6802
Coax ctrl
Coax GND
ANT GND
(CC) BY-NC-SA

Terminator Box

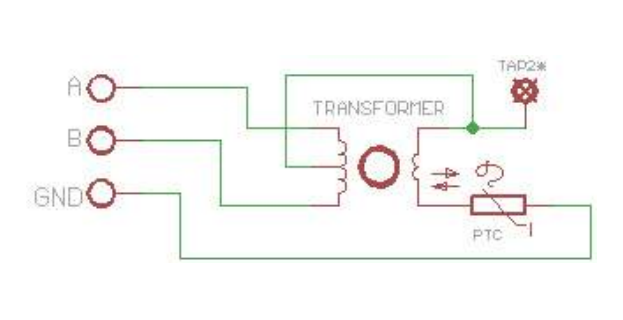
This box contains of Load transformer and protection.

Bi-Directional TWO-WIRE BEVERAGE
TERMINATOR
1,5 - 2,5 m
10 - 20 cm
0,3 - 1 mm
50 Ω
A
B
GND

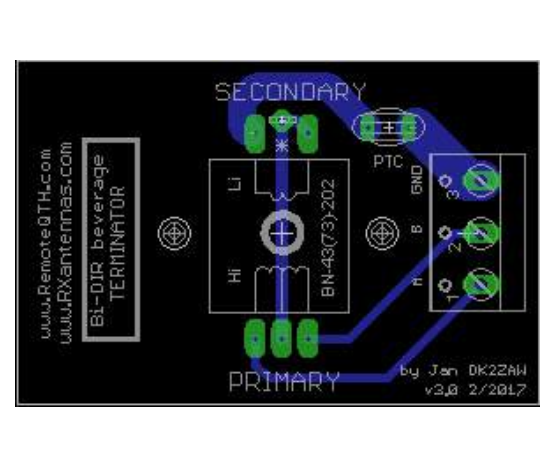
Part list

NAME	PART	VALUE
PTC	PTC	PTC 100mA
Transformer	transformer	BN6802-73

Schematic

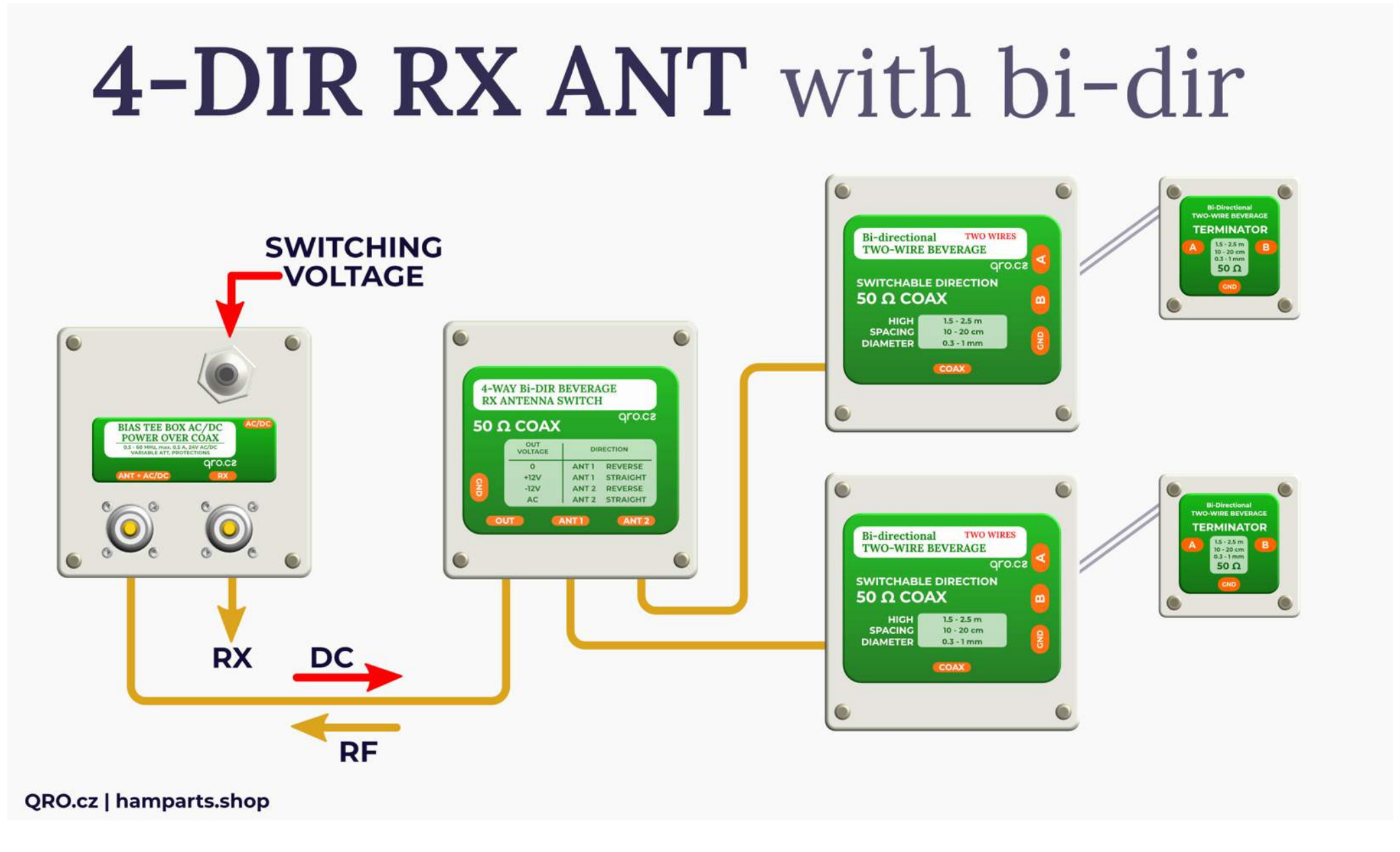

A
B
GND
TRANSFORMER
TAP2
PTC

PCB

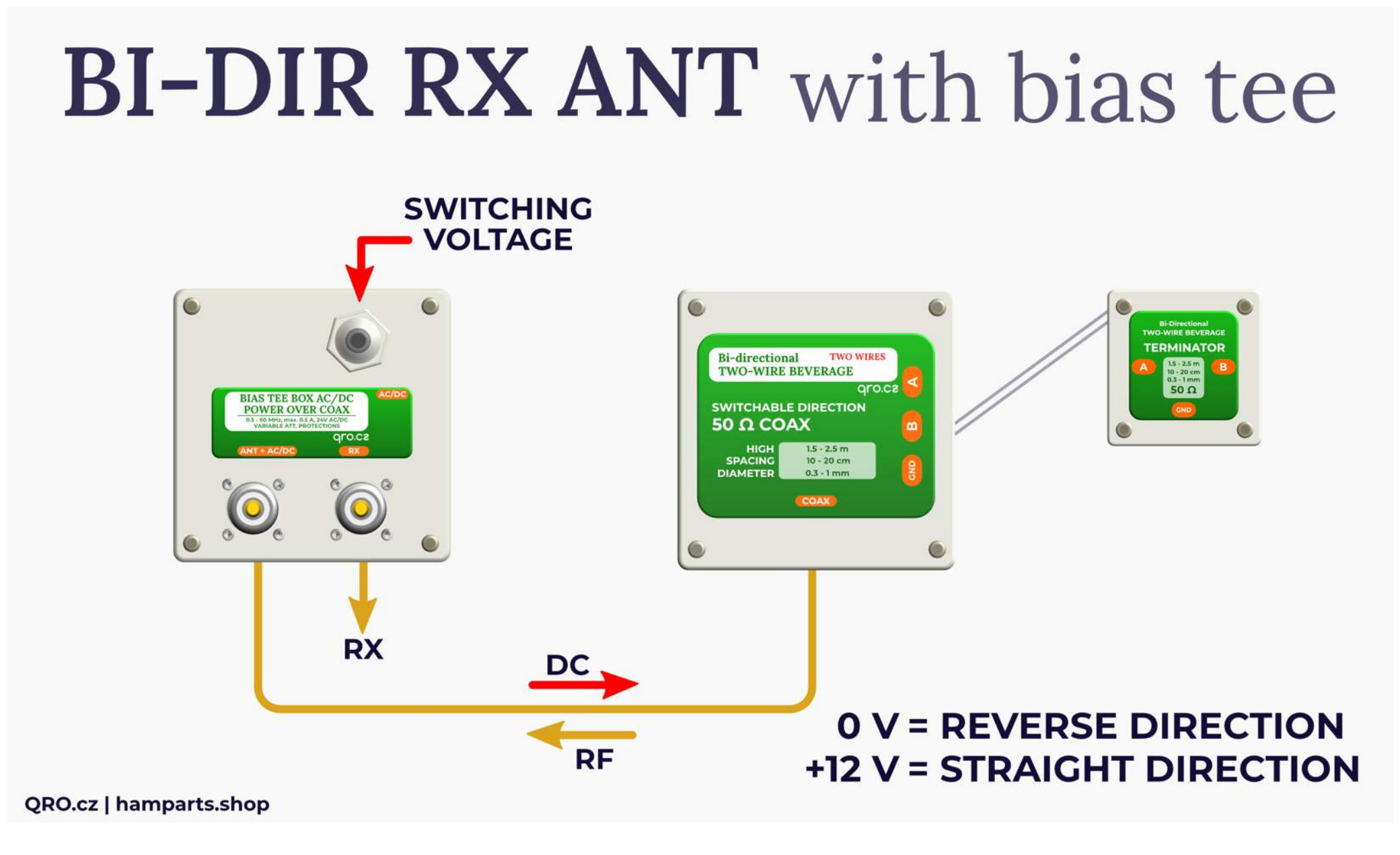

SECONDARY
PRIMARY
www.RemoteQTH.com
www.RXantennas.com
Bi-DIR beverage TERMINATOR
PTC
by Jan DK2ZAH v3.0 2/2017

Applications

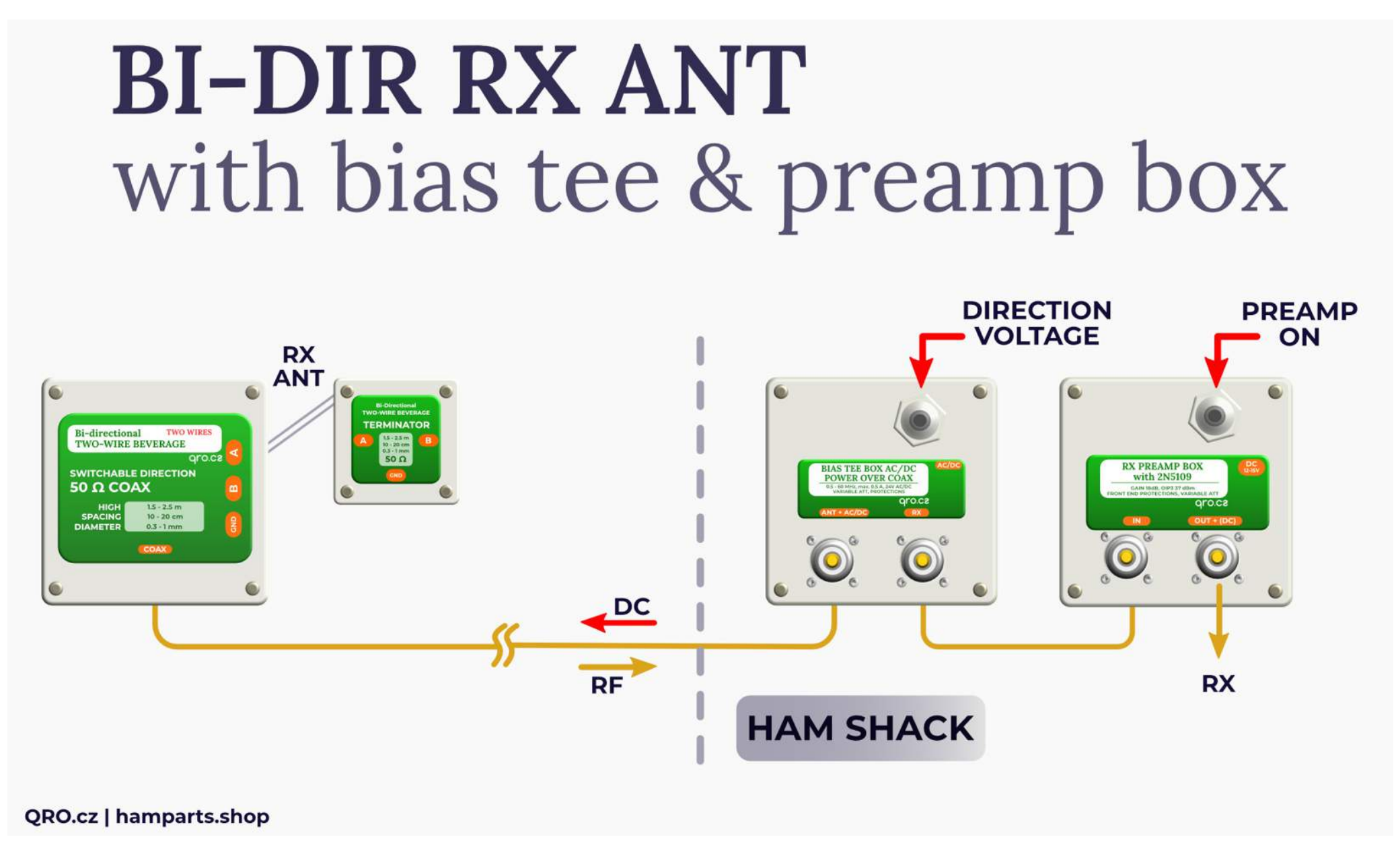
example with **4-way Bi-Dir beverage** and **Bias Tee Box**


4-DIR RX ANT with bi-dir
SWITCHING VOLTAGE
BIAS TEE BOX AC/DC POWER OVER COAX
4-WAY Bi-DIR BEVERAGE RX ANTENNA SWITCH
50 Ω COAX
RX
DC
RF
QRO.cz | hamparts.shop

example with **Bias Tee Box**


BI-DIR RX ANT with bias tee
SWITCHING VOLTAGE
RX
DC
RF
0 V = REVERSE DIRECTION
+12 V = STRAIGHT DIRECTION
QRO.cz | hamparts.shop

example with **Bias Tee Box** and **Preamp Box**


BI-DIR RX ANT with bias tee & preamp box
RX ANT
DIRECTION VOLTAGE
PREAMP ON
DC
RF
RX
HAM SHACK
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KIT

1 coax version with PL SO-239 connector

1 coax version with F connector

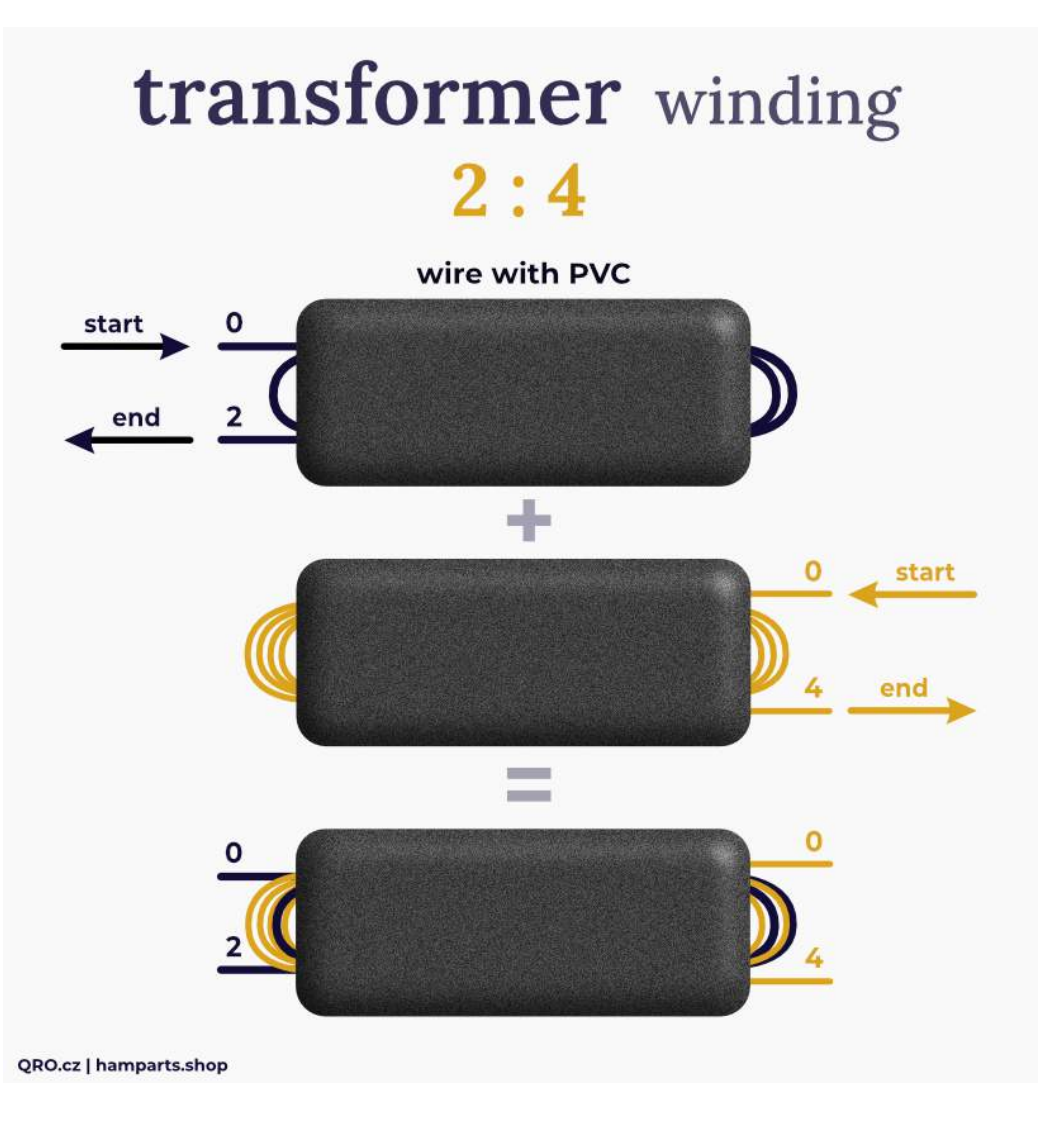
Transformer winding

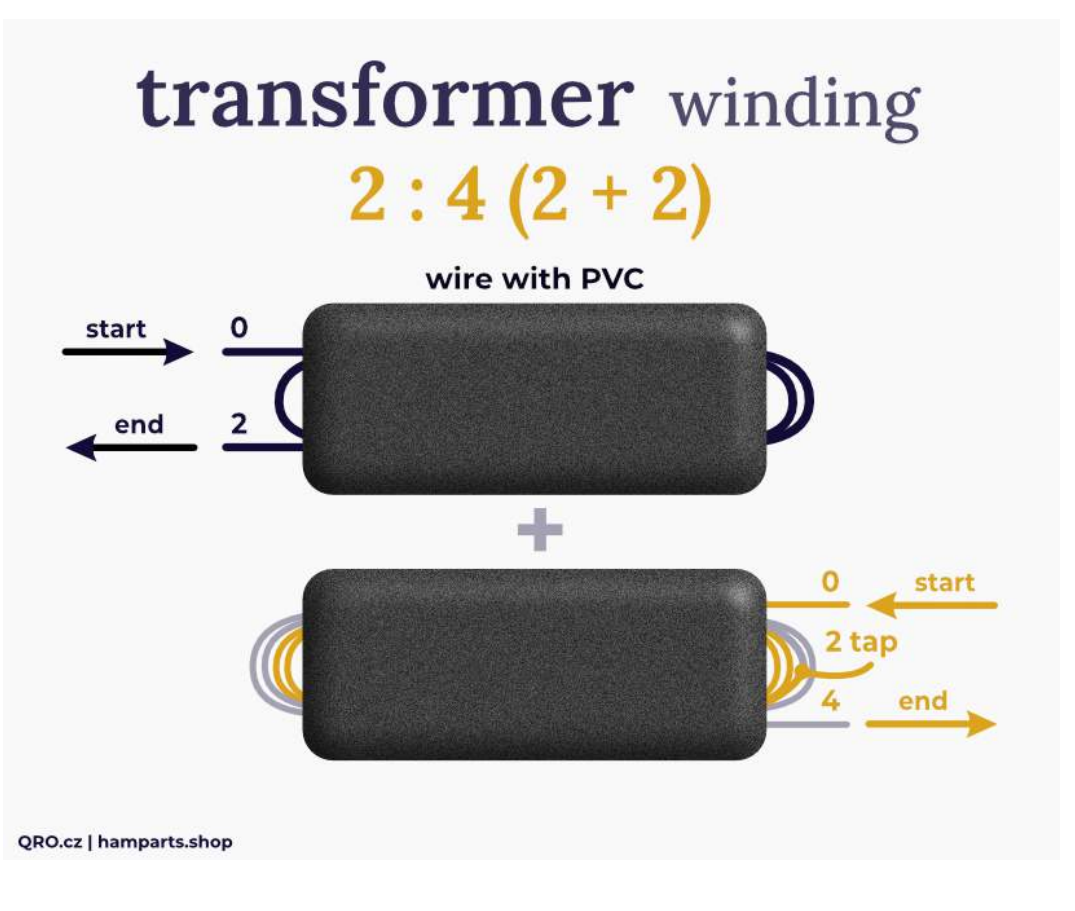
Transformers winding - ⚠ very important thing!

This is very important part of the Bi-Dir antenna. All three transformers are calculated directly for antenna construction (wire diameters, space, height...). We recommend to read more of the theory in ON4UN's LBDX or google.

ℹ Use **our simply TRANSFORMER CALCULATION FILE**.

example of transformer winding


transformer winding
2 : 4
wire with PVC
start
end
0
2
+
0
4
start
end
=
0
2
0
4
QRO.cz | hamparts.shop


transformer winding
2 : 4 (2 + 2)
wire with PVC
start
end
0
2
+
0
start
2 tap
4
end
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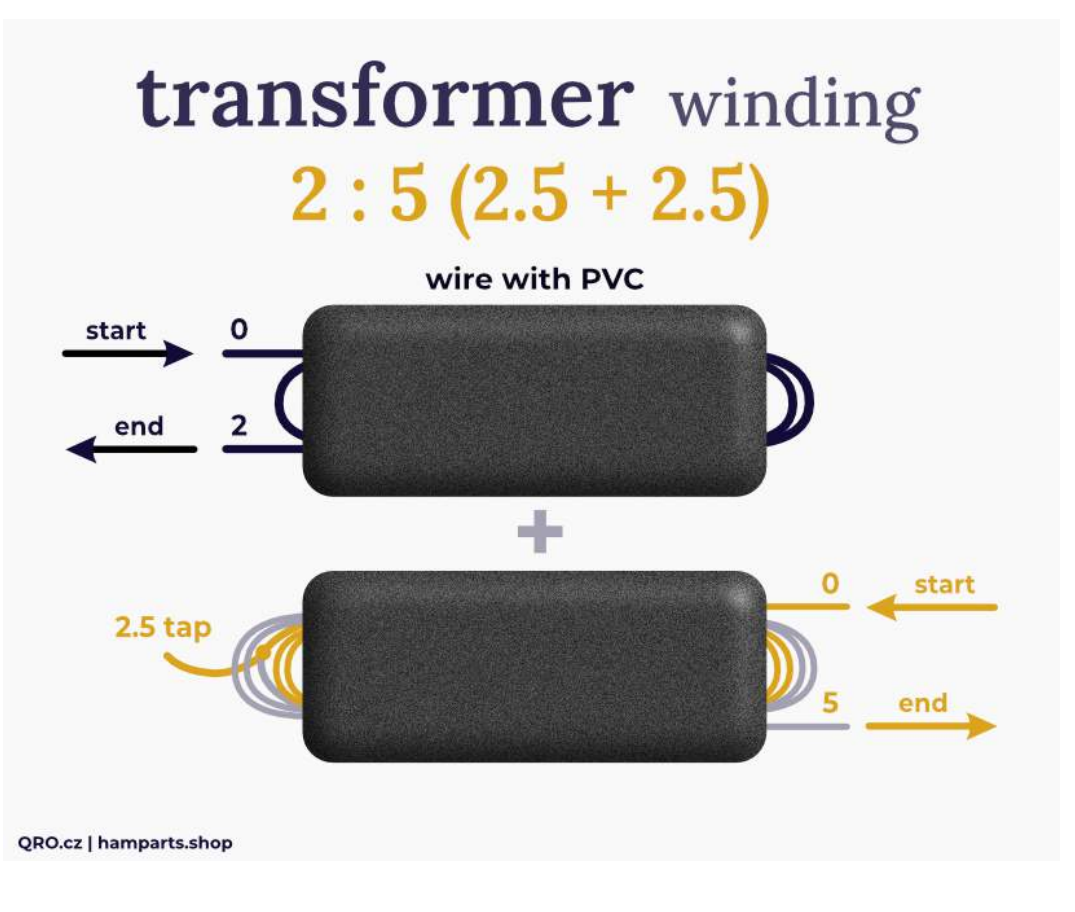
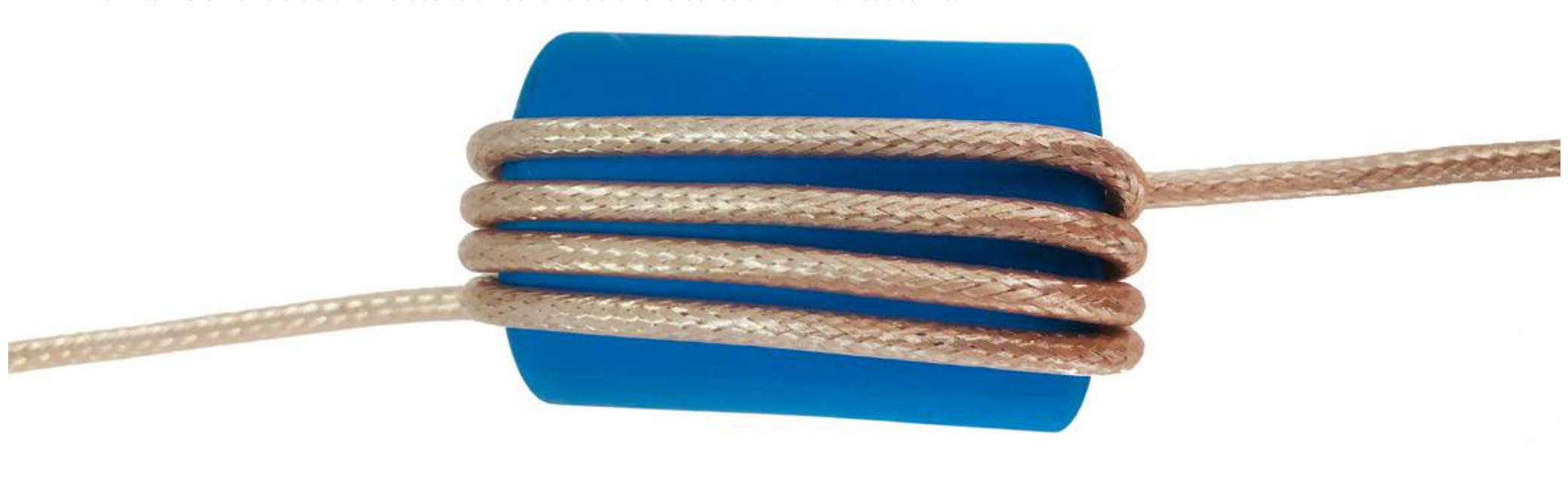
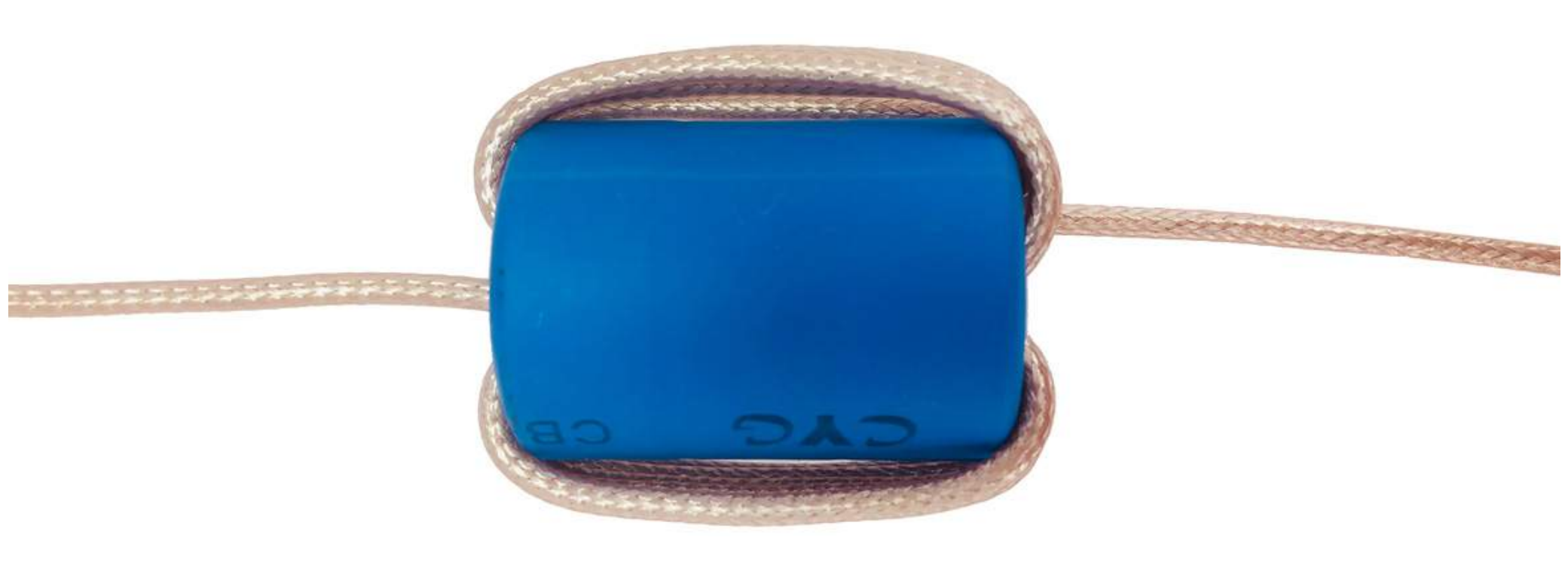
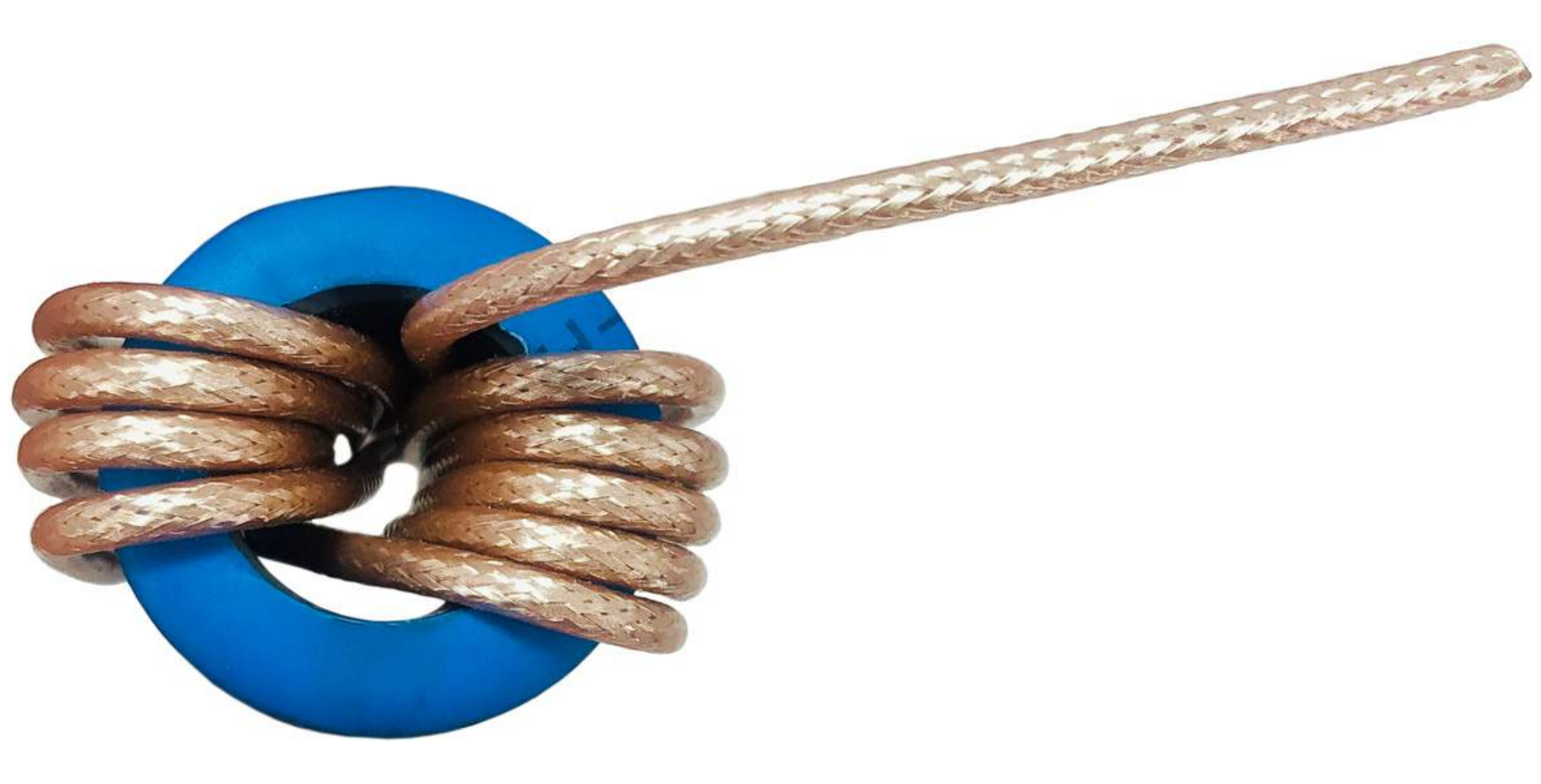

transformer winding
2 : 5 (2.5 + 2.5)
wire with PVC
start
end
0
2
+
2.5 tap
0
start
5
end
QRO.cz | hamparts.shop

CMCC construction

- common-mode current choke construction
- wind 4 turns on one side then cross to another side of the cores and wind rest turns.







Article

ℹ Read **article by Greg Smith, ZL3IX** and article **CLASSIC vs BOG beverage antenna**.

FAQ

− Grounding

- GND screw on feeder box is insulated from Coax shield. This isolation prevents introducing unwanted signals and noise into the antenna system. Grounds are insulated by Common-Mode filter.
- Connect the ground terminal screws only to earth ground using at least one (copper/army GND rods with screw) ground rod. Use as large diameter ground rod as possible.
- A good ground system improves performance (lower noise) and enhances lightning survivability.

− Antenna construction

- The parallel wire line used to construct the antenna must be perfect balanced transmission line conditions.
- The wires must be parallel to the ground. A line should not be constructed in a vertical fashion, using single support poles with one wire under the other. The closer the wire spacing of the line, the lower the impedance of the line, and the less critical installation-related unbalances become.
- When the antenna is ladder-line (300 or 450 Ohm) - twist the ladder line multiple times along its length.
- With advantage you can use **wire antenna insulator** pack which you can find at our web category **PARTS & ACCESSORIES**.

− Switching the direction

- One coax version has got internal relay for direction switching. Without voltage on coax, the antenna works in REVERSE direction.
- You have to apply +12V (or AC) to switch direction to STRAIGHT.
- Please use Bias-Tee or controller which is designed for low bands. Do not use cheap bias-tee boards from China which are usually designed for higher bands (VHF).

− Construction parameters

- Compared to a classic single wire beverage, these Bi-Dir parts are designed exactly for the right type of mechanical/electric construction. For example you can not use 2 wire (10-20cm spaced) boxes for WD-1 wire.
- The same for construction of BOG and antenna in the air. Parameters = Transformers are calculated directly for the design.
- Beverage antennas should be installed more than a wavelength from other large metallic structures, such as vertical antennas, towers, power lines, electric fence wires etc.
- Antenna units can be mounted on trees, wooden pole or fiberglass rods. Do not use metal pipes. Never mount this unit with the connectors facing up.

− First antenna testing

- When the antenna is mounted on place, you can measure SWR. With VNA or antenna analyser. SWR must be lower than 1:2 for 75 Ohm output and 1:1.7 for 50 Ohm version.

− Check your antenna time to time

- Due to storms, lightning and animals we recommend to check SWR of the antenna time to time. If there is any problem with high SWR, please bring the feeder and terminator boxes close to each other and connect it with three short wires, measure SWR again.

− Signal Strength

- **STRAIGHT DIRECTION**: There is little that can go wrong to reduce the level of the wanted forward signal significantly. Attenuation along a lossy line may have a small effect, and a poor ground connection at the feed end may reduce the signal by a dB or two, since the ground loss forms a potential divider with the common mode characteristic impedance.
- **REVERSE DIRECTION**: The level of the reverse direction signal will be affected more by line attenuation than that of the forward direction signal, since the signal has to traverse the line twice. Lines with a significant loss, such as some samples of WD1, may make a noticeable difference. Any loss in the ground connection at the reflection transformer forms a potential divider with the common mode line impedance, and can also contribute to signal loss.

Product details

Included in package

Bi-directional box assembled and tested or all parts for KIT version
Bi-directional terminator assembled and tested or all parts for KIT version

Box dimensions (mm)	100x100x50 mm - PU box IP56
Box dimensions (mm)	65x58x35 mm - PU box IP56
PCB dimensions (mm)	95x40 mm
PCB dimensions (mm)	50x33 mm
Weight brutto (kg)	0.47 to 0.49 kg

Manufacturer:

Ing. Jan Šustr
ID 05476356, VAT CZ8407024780
Palachova 1777/7, 591 01 Žďár nad Sázavou, Czech republic, Europe

General instructions:

Please also read general instructions before handling the product.

European Union Declaration of Conformity

Manufacturer Ing. Jan Šustr (QRO.cz):

ID 05476356, VAT CZ8407024780
Palachova 1777/7, 591 01 Žďár nad Sázavou, Czech republic, Europe

declares that the assembled product:

BI-DIR 1C/BI-DIR BOG 1C

is compatible with the relevant Union harmonisation legislation directives:
EMC Directive 2014/30/EU
ČSN EN 61000-6-3 ed. 2
ČSN EN 61000-6-1 ed. 2 (333432)

On behalf of Ing. Jan Šustr (QRO.cz)
Ing. Jan Šustr, CEO
1st January 2023

Tag: manual, bi-dir, bi directional, bi-dir 1c, 1coax

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